



Started on	Thursday, 20 August 2026, 8:59 AM
State	Finished
Completed on	Thursday, 20 August 2026, 9:00 AM
Time taken	1 min 1 sec
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void reverse(int n) {
4      int count = 0;
5      int rev = 0, remainder;
6
7      count++;
8      while (n != 0) {
9          count++;
10         remainder = n % 10;
11
12         count++;
13         rev = rev * 10 + remainder;
14
15         count++;
16         n /= 10;
17
18         count++;
19     }
20
21     printf("%d\n", count+2);
22 }
23
24 int main() {
25     int n;
26     if (scanf("%d", &n) == 1 && n > 0) {
27         reverse(n);
28     }
29     return 0;
30 }
31
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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